



- 5 (a) A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

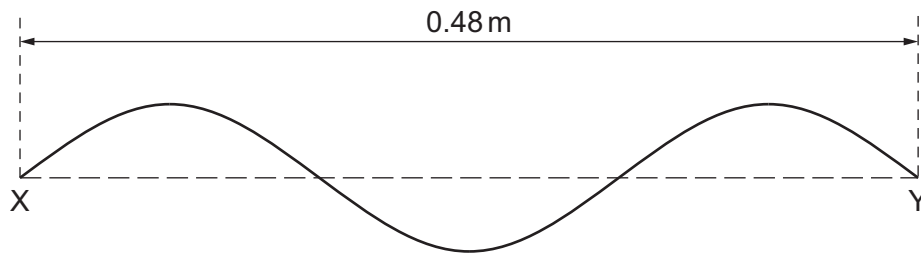


Fig. 5.1

The speed of the wave on the string is 1400 m s^{-1} .

- (i) On Fig. 5.1, draw a cross (×) at **one** position that is a node and another cross at **one** position that is an antinode. Label the node N and the antinode A. [1]
- (ii) Show that the wavelength of the wave produced is 0.32 m. Explain your reasoning. [1]

- (iii) Calculate the frequency of the wave.

frequency = Hz [2]



- (b) A source of sound waves of frequency 780 Hz is on a rotating platform. The speed of the source is 39 m s^{-1} .

The sound is detected by an observer that is a large distance from the rotating platform, as shown in Fig. 5.2.

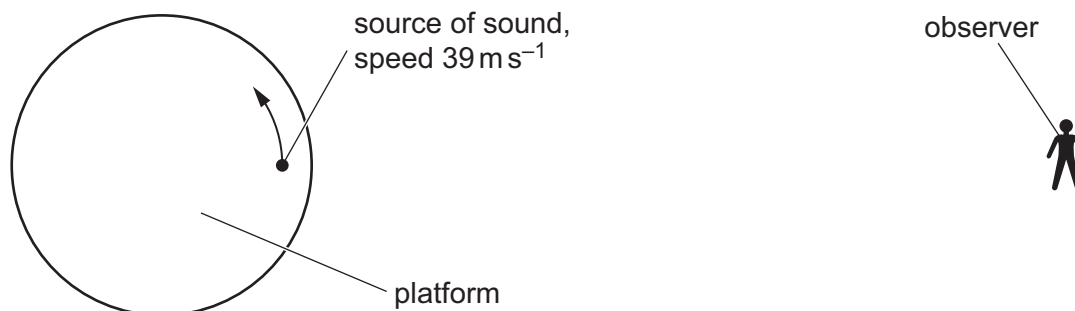


Fig. 5.2 (not to scale)

- (i) The speed of sound in air is 320 m s^{-1} .

Calculate the maximum frequency of the sound detected by the observer.

maximum frequency = Hz [2]

- (ii) At time $t = 0$, the observer detects the sound emitted by the source when it was in the position shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with t of the frequency f of the sound detected by the observer for one complete rotation of the platform. Calculations are not required.



Fig. 5.3

[2]

[Total: 8]

